Experiment 2: Implementing TDMA Slot Allocation in GSM

Objectives, MATLAB Programs and Output Images

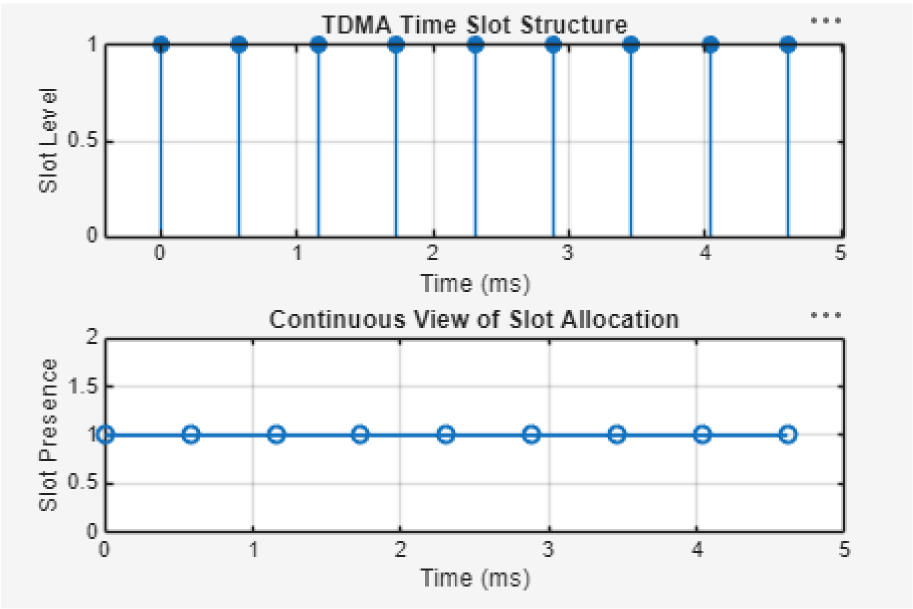
Objective 1

To calculate the Time Slot Duration in a GSM TDMA frame by dividing the total frame time into equal time slots. This helps in understanding the time structure of GSM communication.

MATLAB Program

```
clc; clear; close all;
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;

t = 0:SlotDuration:FrameTime;

figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level')
grid on

subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')

xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```

Output Image – Objective 1



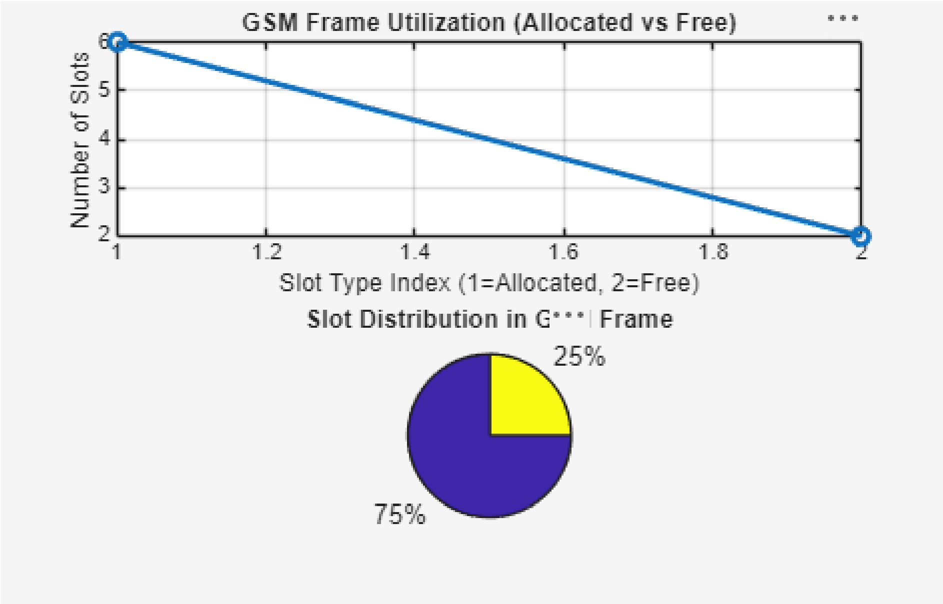
Objective 2

To determine the Frame Utilization (%) based on the number of allocated time slots compared to total available slots. This evaluates the efficiency of TDMA resource usage.

MATLAB Program

```
clc; clear; close all;

TotalSlots = 8;
AllocatedSlots = 6;
FreeSlots = TotalSlots - AllocatedSlots;
Utilization = (AllocatedSlots/TotalSlots)*100;

disp('Frame Utilization (%)')
disp(Utilization)

figure

% Graph 1: Simple Line Plot (SAFE - no bar function)
subplot(2,1,1)
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
title('GSM Frame Utilization (Allocated vs Free)')
xlabel('Slot Type Index (1=Allocated, 2=Free)')
ylabel('Number of Slots')
grid on

% Graph 2: Pie Chart (SAFE)
subplot(2,1,2)
pie([AllocatedSlots FreeSlots])
title('Slot Distribution in GSM Frame')
legend('Allocated','Free')
```

Output Image – Objective 2



Objective 3

To compute the Number of Allocated Time Slots for multiple users in the GSM system. This ensures proper scheduling and interference-free communication.

MATLAB Program

```
clc; clear; close all;

Users = 1:8;             % Total users
TotalSlots = 5;          % Available slots

Alloc = zeros(1,8);    % Initialize allocation
Alloc(1:TotalSlots) = 1; % Assign slots to first users

Blocked = 1 - Alloc;        % Remaining users blocked

disp('User Wise Slot Allocation')
disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'}))

figure

% Graph 1: Allocation per user
subplot(2,1,1)

stem(Users,Alloc,'filled','LineWidth',2)
title('TDMA Slot Allocation per User')
xlabel('User Index')
ylabel('Slot Status (1=Allocated, 0=Blocked)')
ylim([-0.2 1.2])
grid on

% Graph 2: System view (safe alternative to bar)
subplot(2,1,2)
plot(Users,Alloc,'-o','LineWidth',2)
hold on
plot(Users,Blocked,'-s','LineWidth',2)
title('TDMA Slot Utilization Analysis')
xlabel('User Index')
ylabel('Status')
legend('Allocated','Blocked')
grid on
```

The first graph shows which users are allocated time slots and which are blocked due to limited resources.
The second graph compares allocated and blocked users, clearly showing TDMA resource limitation.
Result:

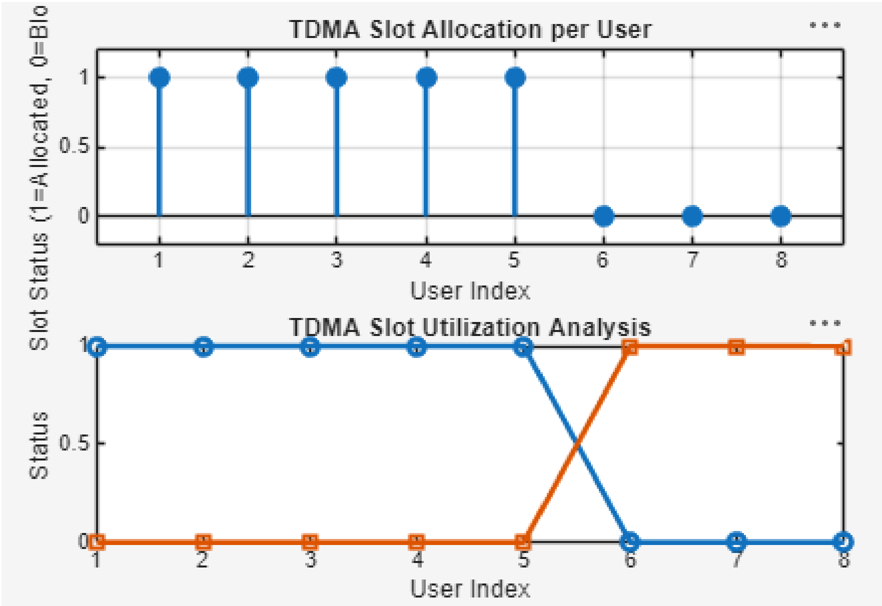
1. The time slot duration in the GSM TDMA frame was successfully calculated, confirming equal division of the frame into fixed time intervals.
2. The frame utilization was effectively analyzed, showing the proportion of allocated and free slots in the TDMA system.

3. The slot allocation per user was successfully implemented, demonstrating that limited slots are assigned to users while excess users are blocked due to resource constraints.

1. Practical Question – 1

A GSM system uses TDMA technique to divide the available channel into multiple time slots for efficient user communication. Using MATLAB, simulate and analyze the TDMA slot

Output Image – Objective 3


TDMA Slot Allocation per User
Slot Status (1=Allocated, 0=Blo
User Index
TDMA Slot Utilization Analysis
Status
User Index